

Predictive vs Generative

Project: Transparent Book Recommender

1. Recall and choose your starting point

Try the recall before opening notes. Then use a different colour or a labelled second line to correct your explanation.

Image and Voice Pipelines

Without opening the old lesson, explain one key idea from Image and Voice Pipelines. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

Feedback Loops

Without opening the old lesson, explain one key idea from Feedback Loops. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

Similarity by Selected Traits

Without opening the old lesson, explain one key idea from Similarity by Selected Traits. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

2. Practise with fading help

Use W2: Compute the distribution for Generative and predictive objectives. Record the answer once; the lesson and workbook refer to the same attempt.

Given inputs · Procedure I chose · Middle steps · Result and check

Given inputs: _____

Procedure I chose: _____

Middle steps: _____

Result and check: _____

The step I completed without help, and the hint I needed

3. Individual reasoning check

Use the worked case for Generative and predictive objectives. Proposed explanation: “Generative models do not predict”. Decide whether this explanation is supported. Point to the step or supplied fact that decides; explain your repair if needed.

My explanation before partner discussion

4. Independent transfer case

Independent case: Analyse a new corpus independently

For prefix The signal is, counts are high 2, low 5, unknown 3. Compute all probabilities and the most frequent continuation. A current measurement reports high. State what the most frequent continuation does and does not establish.

My independent explanation, trace and limits

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

5. Your learning target

A tool retrieves a book description, predicts a genre, then writes a new blurb. Identify each operation and give one distinct failure to check at each stage.

6. Solve the given case

A labels genre; B writes a new blurb; C computes $24/6$.

Distinguish operations.

Show your trace, calculation or evidence

Write or draw your reasoning and evidence.

7. Check a changed case

One model labels a review positive or negative; another writes a new review. Which classifies and which generates? Must generated text be true?

My independent answer and reason

Write or draw your reasoning and evidence.

Week 27 Test and reflect

8. Design a revealing test

Create a changed input, boundary case or missing-information case. Write the expected answer or safe response before testing.

Test record · My evidence

Test record: Changed input or condition

My evidence: _____

Test record: Expected answer or fallback

My evidence: _____

Test record: Observed result

My evidence: _____

Test record: Why the result occurred

My evidence: _____

9. Project checkpoint

Keep scoring predictive and explanations templated or carefully generated and verified.

The project change and evidence I will keep

Write or draw your reasoning and evidence.

10. Self check

I can explain the mechanism. I can show evidence. I can handle a changed case. I can name a limit. Mark each as not yet, with help or independently, and attach supporting work.

Teacher review _____ Next step _____

11. Transfer practice from the original programme

A tool retrieves a book description, predicts a genre, then writes a new blurb. Identify each operation and give one distinct failure to check at each stage.

My independently reasoned answer

Record support used: none, hint or teacher explanation.

12. W1 Identify the objective

Classify these components by target: label a message request/report; continue a prefix with a sequence; retrieve a stored sensor record. Explain why the application may combine them.

My reasoning and evidence

13. W2 Compute the distribution

For The marker is, calculate the three empirical probabilities, their sum and their meaning.

My reasoning and evidence

14. W3 Follow the prefix

After red is selected, the expanded-prefix table gives end 0.8 and today 0.2. Explain why these values cannot be substituted for the red/blue/green probabilities.

My reasoning and evidence

15. W4 Analyse a new corpus independently

For prefix The signal is, counts are high 2, low 5, unknown 3. Compute all probabilities and the most frequent continuation. A current measurement reports high. State what the most frequent continuation does and does not establish.

My reasoning and evidence

16. W5 Design a useful evaluation

A system reports current room status using retrieved measurements and generated sentences. Name checks for evidence, response quality and permission.

My reasoning and evidence

17. Feedback, return check and learning record

Evidence · My record

Evidence: Worked alone / hint / model

My record: _____

Evidence: First step I repaired and why

My record: _____

Evidence: New case checked later; date and result

My record: _____

Evidence: What I can now explain without notes

My record: _____

Evidence: One remaining question

My record: _____

Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

18. Pictorial case practice: Predictive vs Generative

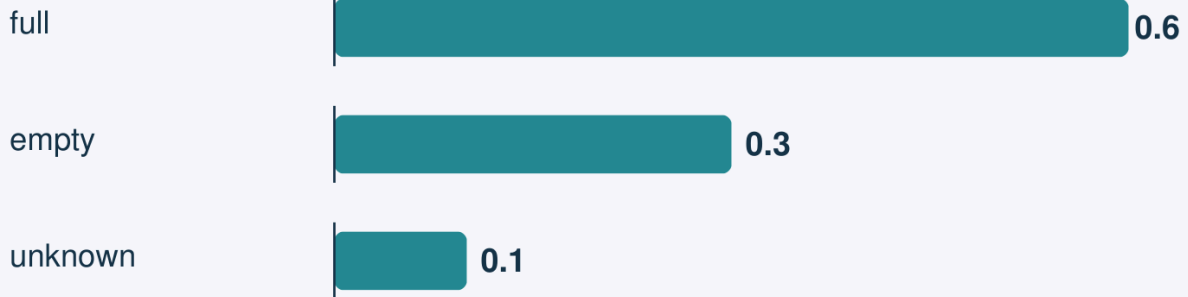
A fictional toy generator predicts the next word after “The box is”.

Observed training continuations: full 6, empty 3, unknown 1.

Use counts divided by total 10. The current physical box is separately checked and found empty.

Toy next-word distribution

probability



Bar length shows the supplied value; the marked baseline is zero.

Predictive vs Generative

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Count table

Continuation counts are 6,3,1; total 10.



2. Probabilities

full=0.6; empty=0.3; unknown=0.1.



3. Greedy continuation

Largest probability chooses full for this toy prefix.



4. Check reality

Current observation says empty; frequent wording is not a present inventory reading.

CHANGE ONE THING

Change the training table to full 2, empty 7, unknown 1.

Explain the changed result using the matching lesson.

Does matching the physical box in the changed table prove the generator measures boxes?

My picture-based explanation and changed-case prediction

Support used: alone / hint / worked example. This record is practice evidence.

Project connection: Use approved book facts for recommender explanations; generated fluency is not a replacement for catalogue evidence.

19. Project portfolio milestone

Project: Transparent Book Recommender

Current milestone: Prepare the demonstration. Gather reproducible instructions, evidence, access checks and remaining limitations.

Use approved book facts for recommender explanations; generated fluency is not a replacement for catalogue evidence.

Evidence to keep · My record

Evidence to keep: This week's input and deciding step

My record: _____

Evidence to keep: Prediction and actual result (or labelled paper trace)

My record: _____

Evidence to keep: One change since the previous checkpoint and why

My record: _____

Evidence to keep: One error or limit and the next test

My record: _____

Explain what this week adds to your project. Reuse last week's record rather than starting again.

20. Supported practice and learning evidence

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

A fictional toy generator predicts the next word after “The box is”.

Practice 1: explain the deciding evidence

Which labelled step matches this detail? Current observation says empty; frequent wording is not a present inventory reading.

1. Probabilities
2. Count table
3. Greedy continuation
4. Check reality

My selected conclusion, deciding fact and explanation

Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

After your attempt, compare with the supported explanation: Check reality Current observation says empty; frequent wording is not a present inventory reading. The table predicts a continuation in a text exercise, not the contents of the current box. Generating the most frequent token is a model-defined decision. A factual inventory statement needs current evidence. The numerical table is intentionally small; real generative systems condition on richer context and do not simply consult these three counts.

Practice 2: explain the deciding evidence

Changed-case practice: Change the training table to full 2, empty 7, unknown 1. Which conclusion is supported by the supplied facts?

1. Greedy output becomes empty with probability 0.7.
2. This one example guarantees correct results for every future input.
3. We can decide the result without examining the changed input or the procedure.

My selected conclusion, deciding fact and explanation

Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

After your attempt, compare with the supported explanation: Greedy output becomes empty with probability 0.7. The continuation distribution changed; the independently observed box has not changed.

Repair a missing building block

Review Image and Voice Pipelines (ai-2/26). Goal: Explain the mechanism and terminology of speech pipelines and multimodal evidence. Short practice: Which stage first fails, and why should the faulty transcript not be silently overwritten?. Return to this week and explain what you changed.

Review Feedback Loops (ai-2/24). Goal: Implement a deterministic recommender. Short practice: What extra field would reveal why comparing raw clicks is misleading?. Return to this week and explain what you changed.

Review Similarity by Selected Traits (ai-2/21). Goal: Represent items with numeric features. Short practice: Explain why B ceases to be a closest option without any candidate data changing.. Return to this week and explain what you changed.

Keep your completed work

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run evidence. Open the HTML copy to read or print it. Drafts stay in this browser when storage is available; clear drafts on a shared device.

What I repaired, the evidence I kept and the next test
